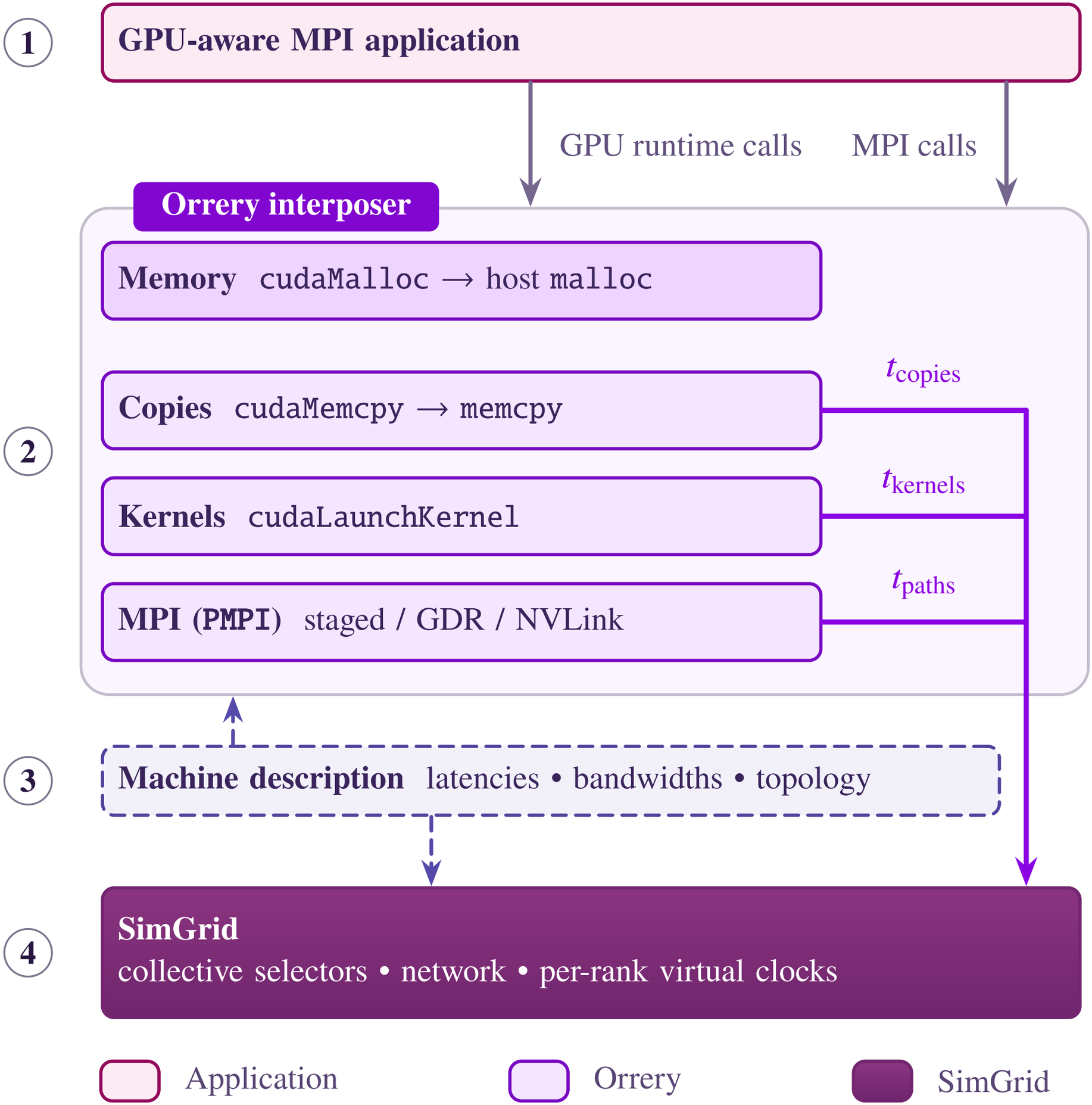

1
GPU-aware MPI application
GPU runtime calls
MPI calls
Orrery interposer
Memory cudaMalloc → host malloc
Copies cudaMemcpy → memcpy
Kernels cudaLaunchKernel
MPI (PMPI) staged / GDR / NVLink
2
t_{copies}
$t_{kernels}$
t_{paths}
3
Machine description latencies • bandwidths • topology
4
SimGrid
collective selectors • network • per-rank virtual clocks
Application
Orrery
SimGrid